CPE 381 Problems in Laplace Transform

Instructor: Rahul Bhadani

1 Second-Order System Analysis

A system is described by the differential equation:

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + 2x(t)$$

(a) Find the transfer function $H(s) = \frac{Y(s)}{X(s)}$

(b) Determine all poles and zeros of the system

(c) Find the impulse response $h(t)$ for the causal system

(d) Analyze the stability of the causal system

(e) Find the step response of the system

Solution:

(a) Taking Laplace transform of both sides (assuming zero initial conditions):

$$\begin{aligned} s^2Y(s) + 5sY(s) + 6Y(s) &= sX(s) + 2X(s) \\ (s^2 + 5s + 6)Y(s) &= (s+2)X(s) \\ H(s) = \frac{Y(s)}{X(s)} &= \frac{s+2}{s^2+5s+6} \end{aligned}$$

(b) Factoring numerator and denominator:

$$H(s) = \frac{s+2}{(s+2)(s+3)}$$

Zero: $s = -2$

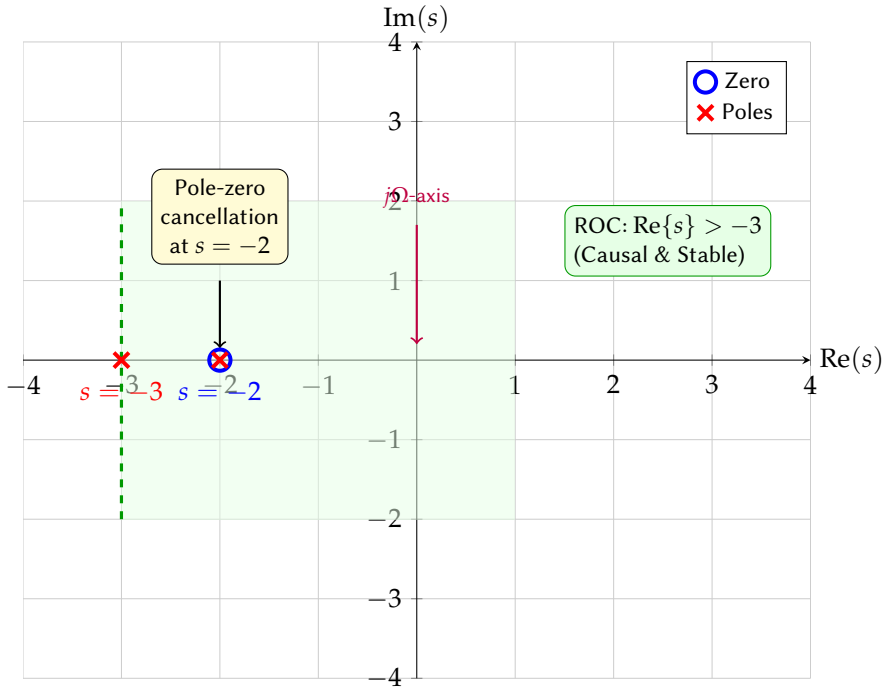
Poles: $s = -2, -3$ (Note: pole-zero cancellation at $s = -2$)

(c) After pole-zero cancellation:

$$H(s) = \frac{1}{s+3}, \quad \text{Region of Convergence, ROC: } \mathrm{Re}\{s\} > -3$$

$$h(t) = e^{-3t}u(t)$$

(d) The system is stable because the only pole is at $s = -3$ (in Left Hand Plane (LHP)) and the ROC includes the $j\Omega$-axis.



(e) Step response:

$$Y(s) = H(s)X(s) = \frac{1}{s+3} \cdot \frac{1}{s} = \frac{1}{s(s+3)}$$
$$= \frac{A}{s} + \frac{B}{s+3}$$

Doing cross multiplication for RHS

$$Y(s) = \frac{As + 3A + Bs}{s(s+3)}$$

Equation coefficients of s both sides

$$A + B = 0$$
$$3A = 1 \Rightarrow A = \frac{1}{3}, \quad B = -A = -\frac{1}{3}$$

Hence,

$$Y(s) = \frac{1}{3s} - \frac{1}{3}\frac{1}{s+3}$$

Using Laplace transform table,

$$y(t) = \frac{1}{3}u(t) - \frac{1}{3}e^{-3t}u(t)$$

2 System Stability Analysis

Given the transfer function:

$$H(s) = \frac{s^2 + 4s + 13}{(s+1)(s^2 + 2s + 5)}$$

(a) Find all poles and zeros

(b) Determine the possible ROCs and corresponding system properties (causal/anti-causal, stable/unstable)

(c) For the causal case, find the impulse response

(d) Check if the system is BIBO stable when causal

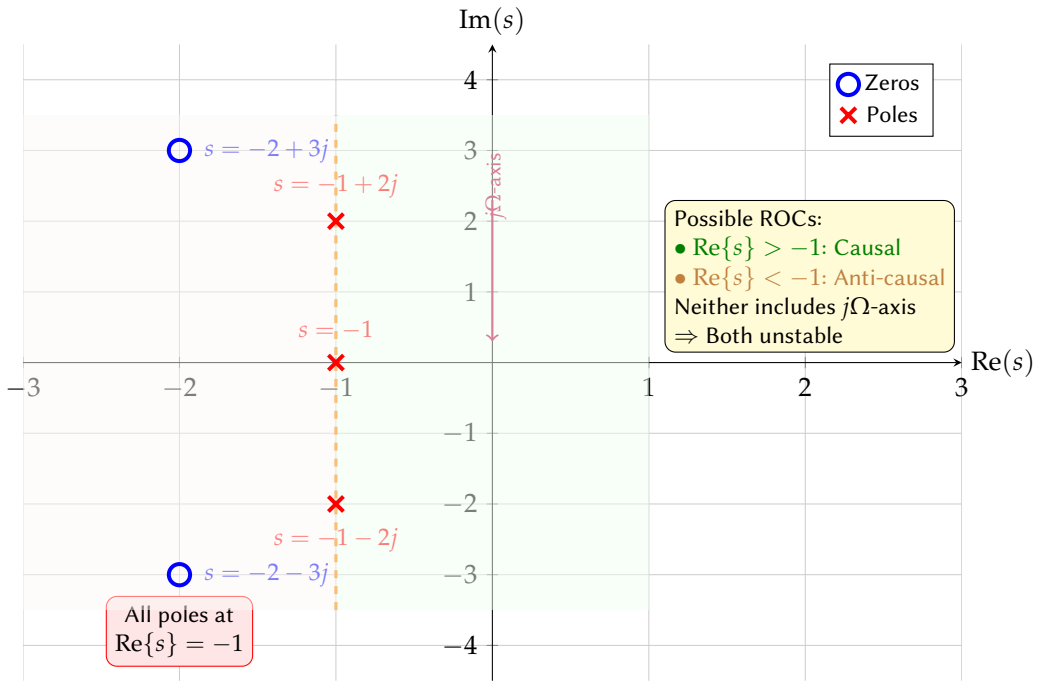
Solution:

(a) Finding poles and zeros:

$$\text{Zeros: } s^2 + 4s + 13 = 0 \Rightarrow s = -2 \pm 3j$$
$$\text{Poles: } s = -1, \quad s^2 + 2s + 5 = 0 \Rightarrow s = -1 \pm 2j$$

(b) Possible ROCs:

- $\text{Re}\{s\} > -1$: Causal, stable (ROC includes $j\Omega$-axis)
- $-1 < \text{Re}\{s\} < -1$: Not possible (poles at same real part)
- $\text{Re}\{s\} < -1$: Anti-causal, unstable



(c) For causal case ($\text{Re}\{s\} > -1$):

$$H(s) = \frac{s^2 + 4s + 13}{(s+1)(s^2 + 2s + 5)} = \frac{A}{s+1} + \frac{Bs + C}{s^2 + 2s + 5}$$

Solving for coefficients:

$$H(s) = \frac{As^2 + 2sA + 5A + Bs^2 + Cs + Bs + C}{(s+1)(s^2+2s+5)}$$
$$A + B = 1 \Rightarrow B = 1 - A$$
$$5A + C = 13 \Rightarrow C = 13 - 5A$$
$$2A + B + C = 4 \Rightarrow 2A + 1 - A + 13 - 5A = 4$$
$$\Rightarrow -4A = -10 \Rightarrow A = 2.5$$
$$B = 1 - 2.5 = -1.5, \quad C = 13 - 5 \times \frac{5}{2} = 13 - 12.5 = 0.5$$

Therefore:

$$H(s) = \frac{2.5}{s+1} + \frac{-1.5s + 0.5}{s^2+2s+5}$$
$$= \frac{2.5}{s+1} - 1.5 \cdot \frac{s+1}{(s+1)^2+4} + 2 \cdot \frac{1}{(s+1)^2+4} = \frac{2.5}{s+1} - 1.5 \cdot \frac{s+1}{(s+1)^2+2^2} + \cdot \frac{2}{(s+1)^2+2^2}$$

Taking inverse Laplace transform using Tables:

$$h(t) = [2.5e^{-t} - 1.5e^{-t}\cos(2t) + e^{-t}\sin(2t)]u(t)$$

(d) The causal system is stable because the ROC ($\text{Re}\{s\} > -1$) includes the $j\Omega$-axis, and all poles are in left-half plane.

3 Mass-Spring-Damper System

A mass-spring-damper system is described by:

$$m\frac{d^2y(t)}{dt^2} + b\frac{dy(t)}{dt} + ky(t) = x(t)$$

where $m = 1$ kg, $b = 4$ Ns/m, $k = 13$ N/m, and $x(t)$ is the applied force.

(a) Find the transfer function

(b) Determine the poles and characterize the system response (overdamped, underdamped, or critically damped)

(c) Find the impulse response

(d) Find the response to a unit step input

Solution:

(a) With given values:

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 13y(t) = x(t)$$
$$s^2Y(s) + 4sY(s) + 13Y(s) = X(s)$$
$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2+4s+13}$$

(b) Poles:

$$s^2 + 4s + 13 = 0 \Rightarrow s = -2 \pm 3j$$

Complex conjugate poles with negative real parts indicate an underdamped system.

(c) Impulse response:

$$H(s) = \frac{1}{(s+2)^2+9} = \frac{1}{3} \cdot \frac{3}{(s+2)^2+3^2}$$
$$h(t) = \frac{1}{3}e^{-2t}\sin(3t)u(t)$$

(d) Step response:

$$Y(s) = H(s)X(s) = \frac{1}{s(s^2+4s+13)}$$
$$= \frac{A}{s} + \frac{Bs+C}{s^2+4s+13}$$

$$\frac{1}{s(s^2+4s+13)} = \frac{As^2+4sA+13A+Bs^2+Cs}{s(s^2+4s+13)}$$
$$\Rightarrow 13A = 1 \Rightarrow A = \frac{1}{13}$$
$$A + B = 0 \Rightarrow B = -A = -\frac{1}{13}$$
$$4A + C = 0 \Rightarrow C = -4A = -\frac{4}{13}$$

Therefore:

$$Y(s) = \frac{1}{13} \cdot \frac{1}{s} - \frac{1}{13} \cdot \frac{s+4}{s^2+4s+13}$$
$$= \frac{1}{13} \cdot \frac{1}{s} - \frac{1}{13} \cdot \frac{s+2}{(s+2)^2+9} - \frac{2}{39} \cdot \frac{3}{(s+2)^2+9}$$

Taking inverse Laplace transform:

$$y(t) = \frac{1}{13}[1 - e^{-2t}\cos(3t) - \frac{2}{3}e^{-2t}\sin(3t)]u(t)$$

4 RLC Circuit Analysis

An RLC circuit has the transfer function:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{s}{s^2+2s+2}$$

(a) Find the differential equation relating $v_{out}(t)$ and $v_{in}(t)$

(b) Determine all poles and zeros

(c) Find the impulse response for the causal system

(d) Is the system stable? Justify your answer

(e) Find the frequency response $H(j\Omega)$

Solution:

(a) From the transfer function:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{s}{s^2 + 2s + 2}$$
$$(s^2 + 2s + 2)V_{out}(s) = sV_{in}(s)$$

Taking inverse Laplace transform:

$$\frac{d^2 v_{out}(t)}{dt^2} + 2\frac{dv_{out}(t)}{dt} + 2v_{out}(t) = \frac{dv_{in}(t)}{dt}$$

(b) Poles and zeros:

$$\text{Zero: } s = 0$$
$$\text{Poles: } s^2 + 2s + 2 = 0 \Rightarrow s = -1 \pm j$$

(c) Impulse response:

$$H(s) = \frac{s}{(s+1)^2 + 1} = \frac{s+1}{(s+1)^2 + 1} - \frac{1}{(s+1)^2 + 1}$$
$$h(t) = [e^{-t}\cos(t) - e^{-t}\sin(t)]u(t)$$

(d) The system is stable because both poles are in the LHP ($s = -1 \pm j$) and the ROC for the causal system ($\text{Re}\{s\} > -1$) includes the $j\Omega$-axis.

(e) Frequency response:

$$H(j\Omega) = H(s)\Big|_{s=j\Omega} = \frac{j\Omega}{(j\Omega)^2 + 2j\Omega + 2} = \frac{j\Omega}{-\Omega^2 + 2j\Omega + 2}$$

5 Feedback System Analysis

Consider a feedback system with forward path transfer function $G(s) = \frac{1}{s+2}$ and feedback path $H(s) = \frac{1}{s+1}$.

(a) Find the overall closed-loop transfer function

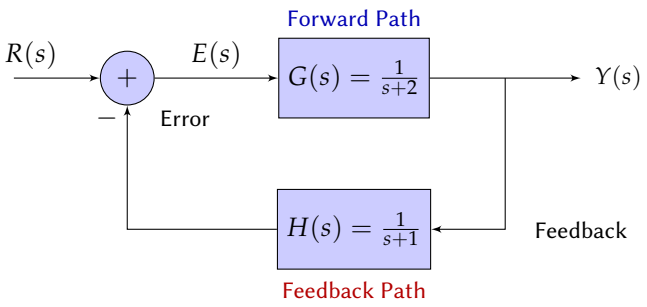
(b) Determine all poles and zeros of the closed-loop system

(c) Analyze the stability of the closed-loop system

(d) Compare the stability with the open-loop system

(e) Find the step response of the closed-loop system

Feedback Control System



Solution:

(a) Closed-loop transfer function:

$$T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{1}{s+2}}{1 + \frac{1}{(s+2)(s+1)}}$$
$$= \frac{\frac{1}{s+2}}{\frac{(s+2)(s+1)+1}{(s+2)(s+1)}} = \frac{1}{s^2 + 3s + 3}$$

(b) Poles and zeros:

Zeros: None

Poles: $s^2 + 3s + 3 = 0 \Rightarrow s = -\frac{3}{2} \pm j\frac{\sqrt{3}}{2}$

(c) The closed-loop system is stable because both poles have negative real parts ($-\frac{3}{2} \pm j\frac{\sqrt{3}}{2}$).

(d) Open-loop transfer function: $G(s)H(s) = \frac{1}{(s+2)(s+1)}$ has poles at $s = -1, -2$ (stable). Both open-loop and closed-loop are stable, but feedback changed the pole locations.

(e) Step response:

$$Y(s) = T(s)X(s) = \frac{1}{s(s^2 + 3s + 3)}$$
$$= \frac{A}{s} + \frac{Bs + C}{s^2 + 3s + 3}$$

$$\frac{1}{s(s^2 + 3s + 3)} = \frac{As^2 + 3As + 3A + Bs^2 + Cs}{s(s^2 + 3s + 3)}$$
$$A = \frac{1}{3}$$
$$A + B = 0 \Rightarrow B = -A = -\frac{1}{3}$$
$$3A + C = 0 \Rightarrow C = -3A = -1$$

Therefore:

$$Y(s) = \frac{1}{3} \cdot \frac{1}{s} - \frac{1}{3} \cdot \frac{s + 3}{s^2 + 3s + 3}$$

Complete the square: $s^2 + 3s + 3 = (s + \frac{3}{2})^2 + (\frac{\sqrt{3}}{2})^2$

$$Y(s) = \frac{1}{3} \cdot \frac{1}{s} - \frac{1}{3} \cdot \frac{s + \frac{3}{2}}{(s + \frac{3}{2})^2 + (\frac{\sqrt{3}}{2})^2} - \frac{1}{2\sqrt{3}} \cdot \frac{\frac{\sqrt{3}}{2}}{(s + \frac{3}{2})^2 + (\frac{\sqrt{3}}{2})^2}$$

Taking inverse Laplace transform:

$$y(t) = \frac{1}{3}[1 - e^{-3t/2}\cos(\frac{\sqrt{3}}{2}t) - \frac{\sqrt{3}}{2}e^{-3t/2}\sin(\frac{\sqrt{3}}{2}t)]u(t)$$

6 Partial Fraction Expansion with Repeated Poles

Given the transfer function:

$$H(s) = \frac{s^2 + 3s + 3}{(s+1)^3}$$

(a) Find the partial fraction expansion

(b) Determine the impulse response $h(t)$

(c) Find the step response

(d) Analyze system stability

Solution:

(a) Partial fraction expansion for repeated poles:

$$\frac{s^2 + 3s + 3}{(s+1)^3} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{(s+1)^3}$$

You can proceed with partial fraction techniques described in previous problems or alternatively you can take a different approach to do partial fraction:

Multiply both sides by $(s+1)^3$:

$$\begin{aligned} s^2 + 3s + 3 &= A(s+1)^2 + B(s+1) + C \\ &= A(s^2 + 2s + 1) + B(s+1) + C \\ &= As^2 + 2As + A + Bs + B + C \\ &= As^2 + (2A + B)s + (A + B + C) \end{aligned}$$

Equating coefficients:

$$\begin{aligned} A &= 1 \\ 2A + B = 3 &\Rightarrow 2(1) + B = 3 \Rightarrow B = 1 \\ A + B + C = 3 &\Rightarrow 1 + 1 + C = 3 \Rightarrow C = 1 \end{aligned}$$

Therefore:

$$H(s) = \frac{1}{s+1} + \frac{1}{(s+1)^2} + \frac{1}{(s+1)^3}$$

(b) Impulse response:

$$h(t) = \left[e^{-t} + te^{-t} + \frac{t^2}{2}e^{-t} \right] u(t)$$

(c) Step response:

$$\begin{aligned} Y(s) = H(s)X(s) &= \left[\frac{1}{s+1} + \frac{1}{(s+1)^2} + \frac{1}{(s+1)^3} \right] \cdot \frac{1}{s} \\ &= \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2} + \frac{D}{(s+1)^3} \end{aligned}$$

$$A = 3, B = -3, C = -2, D = -1$$

Therefore:

$$Y(s) = \frac{3}{s} - \frac{3}{s+1} - \frac{2}{(s+1)^2} - \frac{1}{(s+1)^3}$$

$$y(t) = [3 - 3e^{-t} - 2te^{-t} - \frac{t^2}{2}e^{-t}]u(t)$$

(d) The system is stable because it has a single pole at $s = -1$ (triple pole in LHP).

7 Initial Value Problem with Laplace Transform

Solve the differential equation with initial conditions:

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 4y(t) = e^{-t}u(t), \quad y(0) = 1, \quad y'(0) = 2$$

Solution: Taking Laplace transform of both sides:

$$\begin{aligned}
\mathcal{L}\left\{\frac{d^2y}{dt^2}\right\} &= s^2Y(s) - sy(0) - y'(0) = s^2Y(s) - s - 2 \\
\mathcal{L}\left\{\frac{dy}{dt}\right\} &= sY(s) - y(0) = sY(s) - 1 \\
\mathcal{L}\{y(t)\} &= Y(s) \\
\mathcal{L}\{e^{-t}u(t)\} &= \frac{1}{s+1}
\end{aligned}$$

Substitute into the differential equation:

$$\begin{aligned}
[s^2Y(s) - s - 2] + 4[sY(s) - 1] + 4Y(s) &= \frac{1}{s+1} \\
s^2Y(s) - s - 2 + 4sY(s) - 4 + 4Y(s) &= \frac{1}{s+1} \\
(s^2 + 4s + 4)Y(s) - s - 6 &= \frac{1}{s+1} \\
(s+2)^2Y(s) &= \frac{1}{s+1} + s + 6 \\
Y(s) &= \frac{1}{(s+2)^2(s+1)} + \frac{s+6}{(s+2)^2}
\end{aligned}$$

Solve first term using partial fractions:

$$\frac{1}{(s+2)^2(s+1)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}$$

Multiply both sides by $(s+2)^2(s+1)$:

$$1 = A(s+2)^2 + B(s+1)(s+2) + C(s+1)$$

Substitute $s = -1$: $1 = A(1)^2 \Rightarrow A = 1$
Substitute $s = -2$: $1 = C(-1) \Rightarrow C = -1$
Substitute $s = 0$: $1 = 4A + 2B + C \Rightarrow 1 = 4 + 2B - 1 \Rightarrow 2B = -2 \Rightarrow B = -1$
Therefore:

$$\frac{1}{(s+2)^2(s+1)} = \frac{1}{s+1} - \frac{1}{s+2} - \frac{1}{(s+2)^2}$$

Now solve second term:

$$\frac{s+6}{(s+2)^2} = \frac{D}{s+2} + \frac{E}{(s+2)^2}$$

Multiply both sides by $(s+2)^2$:

$$s + 6 = D(s+2) + E$$

Substitute $s = -2$: $-2 + 6 = E \Rightarrow E = 4$
Equate coefficients: s term: $1 = D$
Therefore:

$$\frac{s+6}{(s+2)^2} = \frac{1}{s+2} + \frac{4}{(s+2)^2}$$

Combine all terms:

$$\begin{aligned} Y(s) &= \left[\frac{1}{s+1} - \frac{1}{s+2} - \frac{1}{(s+2)^2}\right] + \left[\frac{1}{s+2} + \frac{4}{(s+2)^2}\right] \\ &= \frac{1}{s+1} + \frac{3}{(s+2)^2} \end{aligned}$$

Taking inverse Laplace transform:

$$y(t) = e^{-t}u(t) + 3te^{-2t}u(t)$$

8 System Response to Various Inputs

A system has impulse response $h(t) = e^{-2t}u(t)$.

(a) Find the transfer function $H(s)$ and its ROC

(b) Find the response to $x(t) = u(t)$ (unit step)

(c) Find the response to $x(t) = \cos(2t)u(t)$

(d) Find the response to $x(t) = e^{-t}u(t)$

Solution:

(a) Transfer function:

$$H(s) = \frac{1}{s+2}, \quad \text{ROC: } \mathrm{Re}\{s\} > -2$$

(b) Step response:

$$\begin{aligned} Y(s) = H(s)X(s) &= \frac{1}{s+2} \cdot \frac{1}{s} = \frac{1}{s(s+2)} \\ &= \frac{A}{s} + \frac{B}{s+2} \\ A = \frac{1}{2}, \quad B &= -\frac{1}{2} \\ Y(s) &= \frac{1}{2} \cdot \frac{1}{s} - \frac{1}{2} \cdot \frac{1}{s+2} \\ y(t) &= \frac{1}{2}(1 - e^{-2t})u(t) \end{aligned}$$

(c) Response to $\cos(2t)u(t)$:

$$\begin{aligned} X(s) &= \frac{s}{s^2+4} \\ Y(s) = H(s)X(s) &= \frac{1}{s+2} \cdot \frac{s}{s^2+4} \\ &= \frac{A}{s+2} + \frac{Bs+C}{s^2+4} \end{aligned}$$

$$A = -\frac{1}{4}, B = \frac{1}{4}, C = \frac{1}{2} \quad \text{Using Partial Fraction}$$

Therefore:

$$\begin{aligned} Y(s) &= -\frac{1}{4} \cdot \frac{1}{s+2} + \frac{1}{4} \cdot \frac{s}{s^2+4} + \frac{1}{2} \cdot \frac{1}{s^2+4} \\ &= -\frac{1}{4} \cdot \frac{1}{s+2} + \frac{1}{4} \cdot \frac{s}{s^2+4} + \frac{1}{4} \cdot \frac{2}{s^2+4} \\ y(t) &= \left[-\frac{1}{4} e^{-2t} + \frac{1}{4} \cos(2t) + \frac{1}{4} \sin(2t) \right] u(t) \end{aligned}$$

(d) Response to $e^{-t}u(t)$:

$$\begin{aligned} X(s) &= \frac{1}{s+1} \\ Y(s) &= H(s)X(s) = \frac{1}{s+2} \cdot \frac{1}{s+1} = \frac{A}{s+1} + \frac{B}{s+2} \\ A &= \lim_{s \to -1} (s+1)Y(s) = \frac{1}{1} = 1 \\ B &= \lim_{s \to -2} (s+2)Y(s) = \frac{1}{-1} = -1 \\ Y(s) &= \frac{1}{s+1} - \frac{1}{s+2} \\ y(t) &= (e^{-t} - e^{-2t})u(t) \end{aligned}$$

9 Pole-Zero Analysis and System Behavior

Given the transfer function:

$$H(s) = \frac{s^2 - 1}{s^3 + 2s^2 + 2s + 1}$$

(a) Find all poles and zeros

(b) Plot the pole-zero diagram

(c) Determine possible ROCs and corresponding system properties

(d) For each ROC, determine if the system is stable and causal

(e) Find the impulse response for the causal case

Solution:

(a) Poles and zeros:

$$\text{Zeros: } s^2 - 1 = 0 \Rightarrow s = \pm 1$$
$$\text{Poles: } s^3 + 2s^2 + 2s + 1 = 0$$

Try $s = -1$: $(-1)^3 + 2(1) + 2(-1) + 1 = -1 + 2 - 2 + 1 = 0$
So $s = -1$ is a root. Factor:

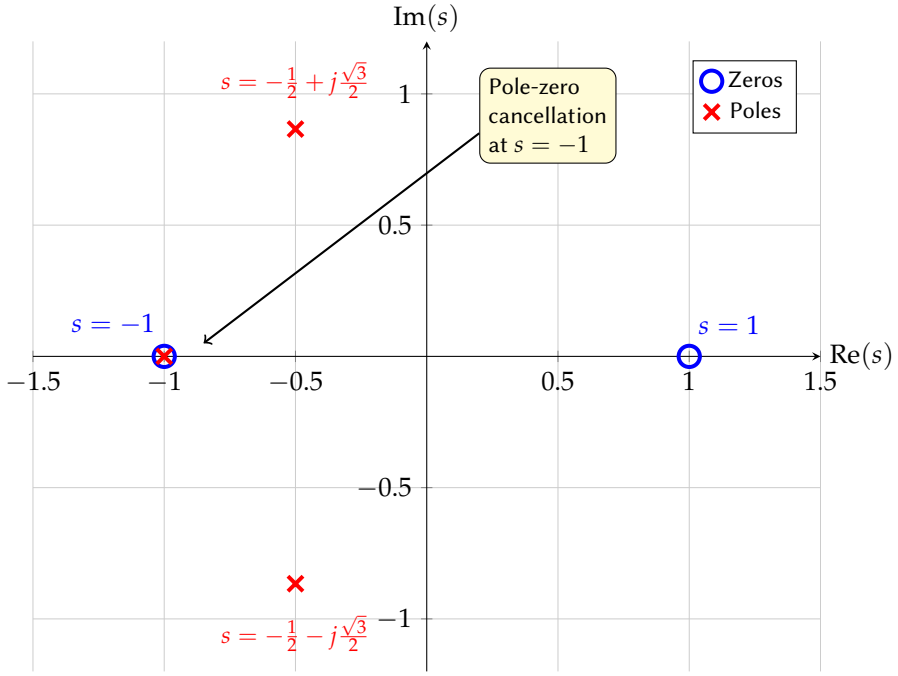
$$s^3 + 2s^2 + 2s + 1 = (s+1)(s^2 + s + 1)$$
$$s^2 + s + 1 = 0 \Rightarrow s = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$$

Poles: $s = -1, s = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$

(b) Pole-zero diagram:

- Zeros: $s = 1$ (right half-plane), $s = -1$ (left half-plane)
- Poles: $s = -1$ (on real axis), $s = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$ (complex conjugate pair in LHP)

Note: There is a pole-zero cancellation at $s = -1$.



(c) After pole-zero cancellation:

$$H(s) = \frac{s-1}{s^2+s+1}$$

Poles: $s = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$
Possible ROCs:

- $\text{Re}\{s\} > -\frac{1}{2}$: Causal
- $\text{Re}\{s\} < -\frac{1}{2}$: Anti-causal

(d) Stability and causality:

- $\text{Re}\{s\} > -\frac{1}{2}$: Causal and stable (ROC includes $j\Omega$-axis)
- $\text{Re}\{s\} < -\frac{1}{2}$: Anti-causal and unstable

(e) Impulse response for causal case:

$$\begin{aligned} H(s) &= \frac{s-1}{s^2+s+1} = \frac{s-1}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} \\ &= \frac{s+\frac{1}{2}}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} - \frac{3}{2}\cdot\frac{1}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} \\ &= \frac{s+\frac{1}{2}}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} - \sqrt{3}\cdot\frac{\frac{\sqrt{3}}{2}}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} \end{aligned}$$

Taking inverse Laplace transform:

$$h(t) = \left[e^{-t/2}\cos(\frac{\sqrt{3}}{2}t) - \sqrt{3}e^{-t/2}\sin(\frac{\sqrt{3}}{2}t)\right]u(t)$$

10 Mixed Causal and Anti-Causal System

Given the Laplace transform:

$$X(s) = \frac{1}{s^2 - 4}$$

(a) Find all possible inverse Laplace transforms by considering different ROCs

(b) For each case, determine if the signal is causal, anti-causal, or two-sided

(c) For each case, determine if the signal is absolutely integrable (BIBO stable)

(d) Find the differential equation that has this as its characteristic equation

Solution:

(a) Partial fraction expansion:

$$X(s) = \frac{1}{s^2 - 4} = \frac{1}{(s-2)(s+2)} = \frac{A}{s-2} + \frac{B}{s+2}$$
$$A = \frac{1}{4}, \quad B = -\frac{1}{4} \quad \text{Using Partial Fraction}$$

Therefore:

$$X(s) = \frac{1}{4} \cdot \frac{1}{s-2} - \frac{1}{4} \cdot \frac{1}{s+2}$$

Possible ROCs and corresponding signals:

- $\text{Re}\{s\} > 2$: $x(t) = \frac{1}{4}[e^{2t} - e^{-2t}]u(t)$ (right-sided)
- $-2 < \text{Re}\{s\} < 2$: $x(t) = \frac{1}{4}[-e^{2t}u(-t) - e^{-2t}u(t)]$ (two-sided)
- $\text{Re}\{s\} < -2$: $x(t) = \frac{1}{4}[-e^{2t} + e^{-2t}]u(-t)$ (left-sided)

(b) Signal properties:

- $\text{Re}\{s\} > 2$: Causal
- $-2 < \text{Re}\{s\} < 2$: Two-sided
- $\text{Re}\{s\} < -2$: Anti-causal

(c) Absolute integrability (BIBO stability):

- $\text{Re}\{s\} > 2$: e^{2t} grows without bound $\Rightarrow$ unstable
- $-2 < \text{Re}\{s\} < 2$: e^{2t} term grows for $t < 0 \Rightarrow$ unstable
- $\text{Re}\{s\} < -2$: e^{-2t} grows for $t < 0 \Rightarrow$ unstable

All cases are unstable.

(d) Differential equation:

$$X(s) = \frac{1}{s^2 - 4} = \frac{Y(s)}{X(s)}$$
$$s^2Y(s) - 4Y(s) = X(s)$$
$$\frac{d^2y(t)}{dt^2} - 4y(t) = x(t)$$

11 System Interconnections

Two systems are connected in cascade:

- System 1: $H_1(s) = \frac{s}{s+1}$ with ROC: $\text{Re}\{s\} > -1$
- System 2: $H_2(s) = \frac{1}{s+2}$ with ROC: $\text{Re}\{s\} > -2$

(a) Find the overall transfer function $H(s) = H_1(s)H_2(s)$

(b) Determine the ROC of the overall system

(c) Find the impulse response of the overall system

(d) Is the overall system stable?

(e) What would be the ROC if the systems were connected in parallel?

Solution:

(a) Overall transfer function:

$$H(s) = H_1(s)H_2(s) = \frac{s}{s+1} \cdot \frac{1}{s+2} = \frac{s}{(s+1)(s+2)}$$

(b) ROC of cascaded system: Intersection of individual ROCs

$$\begin{aligned} \text{ROC}_1: &\quad \text{Re}\{s\} > -1 \\ \text{ROC}_2: &\quad \text{Re}\{s\} > -2 \\ \text{Overall ROC}: &\quad \text{Re}\{s\} > -1 \end{aligned}$$

(c) Impulse response using partial fractions:

$$\begin{aligned} H(s) &= \frac{s}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} \\ A &= -1 \quad \text{Using Partial Fraction} \\ B &= 2 \\ H(s) &= -\frac{1}{s+1} + \frac{2}{s+2}, \quad \text{ROC: } \text{Re}\{s\} > -1 \\ h(t) &= [-e^{-t} + 2e^{-2t}]u(t) \end{aligned}$$

(d) The overall system is stable because the ROC ($\text{Re}\{s\} > -1$) includes the $j\Omega$-axis, and all poles ($s = -1, -2$) are in the LHP.

(e) For parallel connection:

$$\begin{aligned} H(s) = H_1(s) + H_2(s) &= \frac{s}{s+1} + \frac{1}{s+2} \\ &= \frac{s(s+2) + (s+1)}{(s+1)(s+2)} = \frac{s^2 + 2s + s + 1}{(s+1)(s+2)} = \frac{s^2 + 3s + 1}{(s+1)(s+2)} \end{aligned}$$

ROC would still be the intersection: $\text{Re}\{s\} > -1$

12 Final Value Theorem Application

A system has transfer function:

$$H(s) = \frac{2}{s(s+3)}$$

(a) Find the step response $y(t)$

(b) Use the final value theorem to find $\lim_{t\to\infty} y(t)$

(c) Verify your result by taking the limit of $y(t)$ as $t \to \infty$

(d) Under what conditions is the final value theorem valid?

Solution:

(a) Step response:

$$\begin{aligned} Y(s) = H(s)X(s) &= \frac{2}{s(s+3)} \cdot \frac{1}{s} = \frac{2}{s^2(s+3)} \\ &= \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+3} \end{aligned}$$

Using partial fraction

$$\begin{aligned} A &= -\frac{2}{9} \\ B &= \frac{2}{3} \\ C &= \frac{2}{9} \end{aligned}$$

Therefore:

$$\begin{aligned} Y(s) &= -\frac{2}{9} \cdot \frac{1}{s} + \frac{2}{3} \cdot \frac{1}{s^2} + \frac{2}{9} \cdot \frac{1}{s+3} \\ y(t) &= \left[-\frac{2}{9} + \frac{2}{3}t + \frac{2}{9}e^{-3t} \right] u(t) \end{aligned}$$

(b) Final value theorem:

$$\begin{aligned} \lim_{t\to\infty} y(t) &= \lim_{s\to 0} sY(s) = \lim_{s\to 0} s \cdot \frac{2}{s^2(s+3)} \\ &= \lim_{s\to 0} \frac{2}{s(s+3)} = \infty \end{aligned}$$

(c) Verification:

$$\lim_{t\to\infty} y(t) = \lim_{t\to\infty} \left[-\frac{2}{9} + \frac{2}{3}t + \frac{2}{9}e^{-3t} \right] = \infty$$

(d) The final value theorem is valid only if all poles of $sY(s)$ are in the left half-plane. Here, $sY(s) = \frac{2}{s(s+3)}$ has a pole at $s = 0$ (on the imaginary axis), so the theorem gives an incorrect finite value. The actual response goes to infinity due to the double pole at the origin.

13 Relationship Between Differential Equation and Transfer Function

A system is described by the differential equation:

$$\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = 2\frac{dx(t)}{dt} + x(t)$$

(a) Find the transfer function $H(s)$

(b) Find the poles and zeros

(c) Determine the impulse response for the causal system

(d) Find the response to $x(t) = e^{-t}u(t)$ with zero initial conditions

(e) Is the system stable?

Solution:

(a) Taking Laplace transform (zero initial conditions):

$$(s^3 + 6s^2 + 11s + 6)Y(s) = (2s+1)X(s)$$
$$H(s) = \frac{Y(s)}{X(s)} = \frac{2s+1}{s^3 + 6s^2 + 11s + 6}$$

(b) Factor denominator: $s^3 + 6s^2 + 11s + 6 = (s+1)(s+2)(s+3)$
Zeros: $2s + 1 = 0 \Rightarrow s = -\frac{1}{2}$
Poles: $s = -1, -2, -3$

(c) Partial fraction expansion:

$$H(s) = \frac{2s+1}{(s+1)(s+2)(s+3)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3}$$
$$A = -\frac{1}{2}$$
$$B = 3$$
$$C = -\frac{5}{2}$$

Therefore:

$$H(s) = -\frac{1}{2}\cdot\frac{1}{s+1} + 3\cdot\frac{1}{s+2} - \frac{5}{2}\cdot\frac{1}{s+3}, \quad \text{ROC: } \mathrm{Re}\{s\} > -1$$
$$h(t) = \left[-\frac{1}{2}e^{-t} + 3e^{-2t} - \frac{5}{2}e^{-3t}\right]u(t)$$

(d) Response to $x(t) = e^{-t}u(t)$:

$$X(s) = \frac{1}{s+1}$$
$$Y(s) = H(s)X(s) = \frac{2s+1}{(s+1)^2(s+2)(s+3)}$$

This requires partial fraction expansion with repeated poles. The solution would be similar to previous problems but more complex.

(e) The system is stable because all poles ($s = -1, -2, -3$) are in the LHP and the ROC for the causal system ($\mathrm{Re}\{s\} > -1$) includes the $j\Omega$-axis.